Python Stock Analytics Project: Strategy Design and Backtesting

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1. Topic Selected

I chose to work on topic 1 for this analysis because it provides a clear framework for evaluating whether an active, rule-based technical strategy can effectively manage risk and improve performance compared to a passive "Buy-and-Hold" benchmark.

2. System and Data Assumptions

The following parameters were set for the backtesting environment and dataset.

- **Data Source:** Historical market data was retrieved from Yahoo Finance via the yfinance library.
- **Selected Stock Universe:** Five U.S. stocks were selected for analysis: AAPL, MSFT, AMZN, GOOGL, and NVDA.
- **Time Period:** The backtest covers a four-year duration from **January 1, 2020, to January 1, 2024**.
- **Data Frequency:** Daily adjusted closing prices were utilized to account for dividends and stock splits.
- **Starting Capital:** An initial balance of $10,000 was allocated to each stock test.
- **Transaction Logic:** The system assumes a "Long Only" approach with no transaction costs or commissions.
- **Negative Balance Constraint:** As required, a strict logic was implemented to ensure the portfolio balance never dropped below zero.

Table 1: Sample of the raw AAPL dataset downloaded from Yahoo Finance, showing daily price and volume data before strategy calculations.

Price	Open	High	Low	Close	Volume
Date					
2020-01-02	71.344062	72.394093	71.091191	72.333885	135480400
2020-01-03	71.563228	72.389280	71.406689	71.630661	146322800
2020-01-06	70.754006	72.239935	70.503539	72.201401	118387200
2020-01-07	72.211033	72.466315	71.642674	71.861832	108872000
2020-01-08	71.565606	73.318862	71.565606	73.017822	132079200

This dataset was used as the input for calculating moving averages, generating signals, and performing backtesting analysis.

3. Strategy Description

Moving Average Crossover Strategy was implemented in this analysis.

- **Buy Condition:** Enter a position when the **50-day Simple Moving Average (SMA)** crosses above the **200-day SMA** (a "Golden Cross").
- **Sell Condition:** Exit the position and move to cash when the 50-day SMA crosses below the 200-day SMA (a "Death Cross").
- **Objective:** This strategy attempts to capture major upward trends while avoiding severe market downturns by exiting to a cash position.



4. Backtesting Methods

The backtest was developed in Python using pandas, numpy, and matplotlib.

1. **Preprocessing:** Data was cleaned by removing missing values and aligning date indices.
2. **Signal Generation:** Rolling means were calculated, and signals were shifted by one day to avoid "look-ahead bias".
3. **Simulation:** Portfolio value was calculated daily by applying strategy returns to the previous day's capital.
4. **Benchmark:** A "Buy-and-Hold" return was calculated as a primary baseline for comparison.

5. Backtesting Result Analysis

I evaluated performance using total return, annualized return, win rate, and maximum drawdown.

Numerical Summary Table

```
======
BACKTESTING SUMMARY STATISTICS
================================================================================
======
       B&H Total %  B&H Ann %  ...  Strat Win Rate %  Strat Max DD %
AAPL        163.19      27.43  ...             51.95          -26.58
MSFT        142.95      24.90  ...             53.72          -19.44
AMZN         60.10      12.51  ...             50.65          -25.31
GOOGL       104.12      19.57  ...             55.90          -15.66
NVDA        728.90      69.86  ...             54.00          -36.29

[5 rows x 8 columns]
```

Visual Comparisons

AAPL_results image



Buy-and-Hold generates higher returns due to sustained long-term growth, while the moving average strategy reduces volatility and drawdowns by exiting during downturns.

GOOGL_results image



GOOGL is the stock where the strategy outperformed Buy-and-Hold.

NVDA_results image



Here the strategy significantly reduced the maximum drawdown despite having a lower total return.

Key Observations

- **Risk Mitigation:** The strategy significantly outperformed Buy-and-Hold in risk management. In every instance, the Strategy Max Drawdown was lower than the benchmark.
- **Performance vs. Benchmark:** Buy-and-Hold produced higher total returns in 4 out of 5 stocks, except for GOOGL, where the strategy successfully avoided major dips in late 2022.
- **Stability:** The visualizations show that the strategy creates a "smoother" equity curve by staying in cash during sustained market corrections.

6. Conclusion

The analysis demonstrates that the Moving Average Crossover strategy is a more effective tool for capital preservation than for maximizing total returns. While it underperformed the Buy-and-Hold benchmark in hyper-growth scenarios like NVDA, it consistently achieved its goal of reducing extreme drawdowns across all tested stocks. This strategy is best suited for investors prioritizing risk reduction over raw growth.

7. AI Claim

I utilized AI tools ChatGPT and Gemini, to assist with this project. These tools were used to help structure the backtesting Python logic, debug data-alignment issues, and organize the formatting of the final report. All code logic and analytical conclusions were reviewed and verified before final submission.